

```

44.     for field in ['Open', 'High', 'Low', 'Close']:
45.         if field in fields:
46.             data[f'Adj {field}'] = data[field] * data['Adj Ratio']
47.             fields.append(f'Adj {field}')
48.
49.     # Add date features that might be useful
50.     data['Year'] = data.index.year
51.     data['Month'] = data.index.month
52.     data['Day'] = data.index.day
53.     data['Weekday'] = data.index.weekday
54.
55.     # Keep only required fields plus date features
56.     keep_fields = fields + ['Year', 'Month', 'Day', 'Weekday']
57.     data = data[keep_fields]
58.
59.     print(f"Data preparation complete. {len(data)} rows ready for backtesting.")
60.     return data
61.

```

This function handles several important data preparation tasks:

1.
Downloads historical data for the specified ticker and time range
2.
Checks for and handles missing values
3.
Ensures prices are properly adjusted for corporate actions like splits and dividends
4.
Adds date features that might be useful for analysis
5.
Performs basic validation to ensure the data is usable

Avoiding Look-Ahead Bias

One of the most common mistakes in backtesting is look-ahead bias—using information in your strategy that wouldn't have been available at the time of